Implementation Plan

6-DOF Robotic Arm Pick-and-Place and Color Sorting

Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place
Source: IMPLEMENTATION_PLAN.md

Implementation Plan · Robotic Arm Pick-and-Place (Stabilization Focus)

Project: 6-DOF Robotic Arm Pick-and-Place and Color Sorting (current iteration: 4-DOF pick-and-place)
Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place
Report (poster): `24BRS1075_24BRS1065_24BRS1126_Poster_A3 (7).pdf`
Version: 1.2 (pick-and-place stabilization)
Date: 2026-10-22

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1. Purpose

The project report defines the full target system (6-DOF, six MG995 servos, 4 color bins). For this iteration the priority is narrower and explicit: **make the arm move reliably across all joints, make the gripper work properly, prevent mechanical collision/collapse, and complete a stable pick-and-place cycle.** Yellow color sorting and the extra (6th) DOF are deferred until base motion is stable.

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2. Current Focus & Deferred Items

In scope now

- Reliable 4-DOF motion (base, shoulder, elbow, wrist) + gripper.
- Safe, sequential joint motion with per-joint pulse clamping to avoid over-travel.
- Stable grip open/close without servo stalling.
- Complete pick-and-place cycle (approach, grasp, lift, transport, release, home).
- Debounced ultrasonic triggering and R/G/B color sorting.

Deferred (not in this iteration)

- Yellow color detection and 4th bin.
- 6th DOF (wrist roll) servo integration.
- Inverse kinematics and smooth trajectory profiles.

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3. Implementation Phases

Phase 1 · Motion Stabilization (current)

- Add per-joint safe pulse limits and clamping in `pick_and_place` and `integrated_sorting`.
- Replace simultaneous joint updates with sequential updates (`moveArmSafe`) to reduce load/collision.
- Standardize PCA9685 init across sketches.
- Add ultrasonic debounce (3 consecutive readings).
- Keep R/G/B color sorting (Yellow deferred).

Phase 2 · Gripper Reliability

- Define gripper open/close pulses (220/420) with margin to avoid stall.
- Test grip on soft/rigid objects; adjust pulses per hardware.
- Add current-limit awareness (do not command beyond stall).

Phase 3 · Hardware Bring-up (user actions)

- Calibrate each joint with `joint_calibration` and update limits in code.
- Verify power supply (XL4016 set to 5-6 V, ·6 A peak, common ground, 2200 µF cap).
- Secure all horns/screws; check for physical interference before powered motion.
- Run `servo_testing` on each channel individually before integrated motion.

Phase 4 · Validation

- Dry run (no payload) x10 cycles without collision.
- Loaded run (·180 g) x10 cycles.
- Document results in `tests/pick_and_place_test_results.md`.

Phase 5 · Future (backlog)

- Add Yellow sorting and 4th bin.
- Add 6th DOF (wrist roll).
- Implement trapezoidal/S-curve interpolation.
- Develop inverse kinematics.

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4. Change Summary (this iteration)

4.1 Added

- Per-joint safe pulse limits and `clampPulse()` in firmware.
- Sequential joint motion (`moveArmSafe`) to reduce simultaneous load.
- Ultrasonic debounce (3 readings).
- Hardware bring-up checklist (Section 3, Phase 3).

4.2 Changed

- `pick_and_place.ino`: sequential motion, clamped pulses, standardized init.
- `integrated_sorting.ino`: clamped pulses, debounce, standardized init; R/G/B only.
- `color_sensor_test.ino`: reverted to R/G/B classification.
- README status: In Progress (pick-and-place stabilization).
- Docs/tests reverted to 4-DOF R/G/B baseline where previously extended.

4.3 Removed (reverted from prior iteration)

- Wrist roll (6th DOF) channel support.
- Yellow color detection and 4th bin.
- Corresponding test rows (channel 5, Yellow cycle).

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5. Hardware Actions Required (user)

1. **Power:** Set XL4016 output to 5.0-6.0 V; ensure ·6 A peak capability; tie buck GND to Arduino GND; install 2200 µF cap across PCA9685 V+/GND.
2. **Mechanical:** Tighten servo horns and link screws; verify no link hits another during full range; add mechanical stops if needed.
3. **Calibration:** Run `joint_calibration` per joint; record min/max in `tests/servo_test_results.md` and update limits in code.
4. **Gripper:** Test open/close on actual objects; adjust `GRIPPER_OPEN`/`GRIPPER_CLOSED` to avoid stall.

5. **Bring-up order:** `servo_testing` (each channel) · `end_effector_test` · `pick_and_place` (no payload) · loaded test.

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6. Risk & Mitigation

Risk	Mitigation
Servo overload / brownout	Sequential motion, adequate PSU, common ground, capacitor
Mechanical collision	Per-joint limits, calibration, physical stops
Gripper stall	Pulse margin, current monitoring, adjust grip
Misclassification (R/G/B)	Ambient shielding, white balance

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7. Acceptance Criteria

- All 4 joints move within calibrated limits without collision.
- Gripper opens/closes reliably on test objects.
- 10 consecutive no-payload pick-and-place cycles pass.
- 10 loaded cycles (·180 g) pass.
- Results documented.

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8. Deliverables

1. Updated firmware and docs.
2. `IMPLEMENTATION_PLAN.md`.
3. `docs/change_report.pdf`.
4. Git commit and push.